Supplementary Material for “A Reproducible ROI-Robust Evaluation and Mobile Distillation Framework for Ultrasound Lesion Classification”

S1. Result Provenance

This supplement retains only tables tied to the frozen analysis-label snapshot and removes earlier development tables that mixed earlier dataset labels, current XML labels, and intermediate analysis labels. All quantitative tables retained here use the snapshot below:

> `eval_reports/paper_log_label_reconstruction_20260505`
> `/paper_log_case_labels.csv`
> SHA256: `4C5BFFE475B89BA9C52AE1C19FD6558F2C6C31247`
> `19807828C936C1A18139BB1`.

The analysis-label counts are TN5000 1435/3565 benign/malignant, BUSI 435/212, and AUL 183/452. These counts are provenance for the reported analyses and do not overwrite the descriptive dataset-statistics counts in the main manuscript.

Table 1: Primary result sources used by the current manuscript. Internal run directories are archived in the reproducibility package; the manuscript reports functional model names rather than legacy experiment codes.

Evidence block	Archived source family	Notes
Dataset-specific benchmark	Manifest-locked main reruns	TN5000 3 seeds; BUSI/AUL 5 folds
TransXNet-family module sweep	Manifest-locked TN5000 module reruns	8 MCA/MUDD/DA combinations
ROI robustness	Manifest-locked box-noise and detector-ROI report	Same label CSV in run config
Mobile export/runtime	Manifest-locked TN5000 mobile export and two-device Android reports	Re-exported artifacts
Cross-organ probe	EfficientFormer joint/LODO aggregate report	3 seeds
BUSI duplicate audit	Frozen-manifest duplicate audit	Descriptive only

Table 2: Manifest-locked dataset-specific benchmark context. Values are mean ± standard deviation over TN5000 seeds or BUSI/AUL training folds evaluated on the fixed held-out test sets.

Dataset	Method	AUC	BalAcc	F1	Acc
TN5000	UL-TransXNet	0.9537 ± 0.0015	0.8865 ± 0.0064	0.8750 ± 0.0059	0.8960 ± 0.0075
TN5000	ConvNeXt-Tiny	0.9508 ± 0.0020	0.8794 ± 0.0023	0.8731 ± 0.0066	0.8957 ± 0.0065
TN5000	Original TransXNet	0.9467 ± 0.0009	0.8727 ± 0.0023	0.8578 ± 0.0120	0.8803 ± 0.0137
BUSI	UL-TransXNet	0.8494 ± 0.0039	0.7702 ± 0.0076	0.7479 ± 0.0283	0.7752 ± 0.0383
BUSI	Swin-T	0.8190 ± 0.0091	0.7137 ± 0.0095	0.6896 ± 0.0300	0.7194 ± 0.0445
BUSI	Original TransXNet	0.8136 ± 0.0239	0.7531 ± 0.0303	0.7441 ± 0.0292	0.7814 ± 0.0304
AUL	UL-TransXNet	0.8146 ± 0.0165	0.7611 ± 0.0283	0.7589 ± 0.0307	0.7827 ± 0.0329
AUL	Swin-T	0.8319 ± 0.0117	0.7783 ± 0.0325	0.7589 ± 0.0243	0.7717 ± 0.0211
AUL	Original TransXNet	0.7899 ± 0.0090	0.7566 ± 0.0456	0.7547 ± 0.0516	0.7795 ± 0.0539

Table 3: TN5000 TransXNet-family ablation with seed variability. Values are mean ± standard deviation over three seeds.

Variant	AUC	BalAcc	F1	Acc
TransXNet-base	0.9456 ± 0.0006	0.8769 ± 0.0014	0.8526 ± 0.0097	0.8733 ± 0.0107
TransXNet-DA	0.9447 ± 0.0024	0.8785 ± 0.0047	0.8732 ± 0.0075	0.8960 ± 0.0088
TransXNet-MUDD	0.9511 ± 0.0018	0.8857 ± 0.0071	0.8785 ± 0.0019	0.9000 ± 0.0022
TransXNet-MUDD+DA	0.9579 ± 0.0002	0.8901 ± 0.0080	0.8836 ± 0.0074	0.9043 ± 0.0063
TransXNet-MCA	0.9465 ± 0.0046	0.8802 ± 0.0068	0.8663 ± 0.0086	0.8880 ± 0.0085
TransXNet-MCA+DA	0.9441 ± 0.0040	0.8747 ± 0.0078	0.8593 ± 0.0094	0.8817 ± 0.0103
TransXNet-MCA+MUDD	0.9555 ± 0.0011	0.8834 ± 0.0013	0.8757 ± 0.0108	0.8973 ± 0.0119
UL-TransXNet	0.9522 ± 0.0009	0.8754 ± 0.0128	0.8652 ± 0.0004	0.8883 ± 0.0033

Table 4: TN5000 AUC robustness context across the four models shown in the main robustness figure. The run configuration points to the frozen analysis-label CSV listed in Section S1.

Model	GT	5% noise	10% noise	20% noise	Detector
TransXNet-MUDD+DA	0.9623	0.9594	0.9556	0.9463	0.9497
TransXNet-base	0.9469	0.9445	0.9392	0.9292	0.9388
ConvNeXt-Tiny	0.9534	0.9533	0.9524	0.9483	0.9511
Swin-T	0.9506	0.9509	0.9488	0.9463	0.9481

Table 5: Exact TN5000 ROI robustness metrics for the strongest TransXNet-MUDD+DA teacher.

Input box	AUC	BalAcc	Mean box IoU
GT ROI	0.9623	0.9036	1.000
GT ROI + 5% noise	0.9594	0.8969	0.899
GT ROI + 10% noise	0.9556	0.8842	0.810
GT ROI + 20% noise	0.9463	0.8665	0.662
Detector ROI	0.9497	0.8849	0.789

S2. Full Main Benchmark Context

S3. TransXNet-family Design-space Details

S4. ROI Robustness

S5. Android Runtime Details

Table 6: Two-device Android runtime sweep for exported EfficientFormer artifacts. AUC is computed from exported predictions under the frozen analysis-label snapshot. Hot latency is end-to-end total latency in milliseconds.

Device	Artifact	Runtime	Android AUC	Hot total ms	Hot inference ms	P95 total ms
Xiaomi 24129PN74C	ECA+KD	CPU default	0.9552	31.312	30.024	34.023
Xiaomi 24129PN74C	ECA+KD	CPU-4T	0.9552	31.574	30.190	33.191
Xiaomi 24129PN74C	ECA+KD	XNNPACK-4T	0.9552	42.403	40.859	48.883
Xiaomi 24129PN74C	ECA+KD	NNAPI	0.9552	159.080	157.471	179.902
Samsung SM-X800	ECA+KD	CPU default	0.9552	56.507	55.180	58.360
Samsung SM-X800	ECA+KD	CPU-4T	0.9552	56.417	55.090	58.506
Samsung SM-X800	ECA+KD	XNNPACK-4T	0.9552	107.141	105.773	112.055
Samsung SM-X800	ECA+KD	NNAPI	0.9552	279.982	278.047	318.094

Table 7: Exported mobile-student comparison under CPU default runtime. Offline metrics are from the manifest-locked export summary; Android AUC and latency are from the two-device headless run.

Model	Offline AUC	Offline BalAcc	Android AUC	Xiaomi hot ms	Samsung hot ms	Threshold
EfficientFormer-L1+ECA+KD	0.9563	0.8792	0.9552	31.312	56.507	0.87
EfficientFormer-L1+KD	0.9522	0.8699	0.9511	40.364	56.477	0.83
EfficientFormer-L1+ECA	0.9305	0.8400	0.9278	40.517	60.254	0.49

Table 8: EfficientFormer-L1 multi-organ probe. Values are mean ± standard deviation over three seeds.

Protocol	Test domain	AUC	BalAcc
Joint multi-organ	TN5000	0.9212 ± 0.0065	0.8420 ± 0.0044
Joint multi-organ	BUSI	0.8324 ± 0.0192	0.7553 ± 0.0234
Joint multi-organ	AUL	0.7984 ± 0.0367	0.7318 ± 0.0262
LODO zero-shot	TN5000	0.6261 ± 0.0557	0.5723 ± 0.0239
LODO zero-shot	BUSI	0.7339 ± 0.0428	0.6422 ± 0.0415
LODO zero-shot	AUL	0.5936 ± 0.0344	0.5356 ± 0.0572

S6. Multi-organ Joint and LODO Details

S7. BUSI Duplicate Audit

BUSI is retained in its public benchmark form for comparability, but we audited exact and near-duplicate candidates because duplicate or label-conflict images can inflate or distort small public ultrasound benchmarks. The audit below is descriptive; it does not change the main benchmark denominator.

Table 9: Descriptive BUSI duplicate and near-duplicate audit. The audit uses diagnostic labels from the frozen manifest.

Rule	Exact groups	Near-duplicate pairs	Label-conflict pairs	Audit artifact
Strict (≤ 2)	1	88	26	`busi_duplicate_audit_20260510_manifestlabels_threshold2.csv`
Relaxed (≤ 5)	1	141	41	`busi_duplicate_audit_20260510_manifestlabels_threshold5.csv`

S8. Removed Legacy Tables

This submission package intentionally removes earlier supplementary tables for CUDA trade-off, non-locked Android baselines, BUSI/AUL auto-ROI classification, and calibration. Those tables were useful during development, but they were produced before the final manifest discipline was enforced or before the Android threshold/label loader was fixed. They are not used as manuscript evidence. They should only be restored after regeneration from per-case prediction CSVs that reference the label snapshot and manifest hash in Section S1.